
LIMITATIONS OF NEWS SENTIMENT ANALYSIS IN SHORT-TERM STOCK RETURN PREDICTION: A MULTI-LEVEL APPROACH

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ABSTRACT

In this paper, I investigate the challenges and limitations of using news sentiment analysis for predicting next-day stock returns in the U.S. equity market. While sentiment analysis has shown promise in long-term market prediction, its effectiveness for short-term prediction remains contentious. I propose a novel multi-level approach that integrates sentiment analysis across individual stocks, industries, and the broader economy, extending previous hierarchical market analysis frameworks using a comprehensive dataset combining financial news from Bloomberg, Reuters, and Yahoo Finance (2009-2023).

My methodology employs an instruction-pre trained LLaMA 3 8B model for sentiment analysis, selected benchmarking against other state-of-the-art models including RoBERTa Large, FinBERT, and LLaMA 3 8B on the FiQA and FPB datasets. Following recent advances in financial NLP, I augment traditional sentiment features with market data, implementing separate models of high-coverage (>150 articles annually) and low-coverage stocks to account for varying news frequency.

My results demonstrate several key findings. First, even with sophisticated sentiment analysis and feature augmentation, accurate next-day return prediction remains elusive, supporting the efficient market hypothesis at short time horizons. Second, when incorporating both sentiment and market features, models tend to overfit on price-based features, diminishing the impact of sentiment signals. Third, I find that economy-wide sentiment features demonstrate higher predictive power than industry-specific or stock-specific sentiment, suggesting the importance of broader market sentiment over individual stock news.

My trading strategy back tests reveal that the most robust results come from focusing on less-covered stocks with the consistent correlation between sentiment and returns, though even these strategies show limited economic significance when accounting for transaction costs. Surprisingly, sources often considered less reliable (Benzinga, Zacks) showed more stable sentiment signals than traditional news providers. These findings contribute to the growing body of literature on the limitations of sentiment-based trading strategies and highlight the importance of realistic expectations in sentiment-driven algorithmic trading. It should be noted that all analyses and conclusions are subject to the limitations of the underlying datasets used, including potential survivorship bias in the stock universe and varying quality of news coverage across different sources and time periods.

Keywords sentiment analysis · stock return prediction · natural language processing · multi-level analysis · algorithmic trading

1 Introduction

1.1 Background

Financial news and its impact on markets have changed dramatically over the past decade. The rise of digital media has created a complex landscape where news quality and reliability vary significantly across sources. This shift enables

examination of how different news sources affect stock prices, particularly comparing heavily covered stocks against those that receive less media attention Chan [2003].

This research into next-day stock return prediction stems from several key observations. While sentiment analysis works reasonably well for long-term market predictions Baker and Wurgler [2006], its effectiveness for next-day predictions remains unclear. Most studies have looked at either broad market sentiment Tetlock et al. [2008] or individual stock sentiment Loughran and McDonald [2011], but few have tried to connect these different levels of analysis.

Beyond this gap in research, new tools for analyzing sentiment in text through models like LLaMA and BERT Araci [2019] have emerged. Yet these improvements in sentiment analysis haven't led to proportionally better return predictions. This gap raises important questions about how sentiment information actually affects stock prices.

Different types of investors react to news in varying ways. Large institutional investors typically process news quickly in their trading decisions, while retail investors often react more slowly, potentially creating temporary price discrepancies Barber and Odean [2008]. However, as automated trading and sentiment analysis tools become more common, these opportunities may be disappearing.

The study focuses on U.S. stocks from 2009 to 2023, using news from Bloomberg, Reuters, and Yahoo Finance. This comprehensive dataset helps understand how sentiment signals work across different market conditions and news environments. The analysis specifically examines three challenging aspects: how quickly sentiment affects prices, how news quality differs between sources, and how sentiment interacts with traditional market indicators.

1.2 Research objectives

The primary goal is to determine whether news sentiment can effectively predict next-day stock returns. Unlike previous research examining longer time periods, this study focuses specifically on next-day predictions, where market efficiency should be strongest.

A critical objective involves testing four leading language models for financial sentiment analysis: RoBERTa Large, FinBERT, LLaMA 3 8B, and instruction-pretrained LLaMA 3 8B. The models are evaluated on standard financial text datasets (FiQA and FPB) to assess their understanding of financial sentiment.

Another key objective investigates whether combining different types of data improves predictions. The analysis examines traditional market data alongside sentiment indicators at three levels: individual stocks, industries, and the overall economy.

The study also aims to compare prediction quality between stocks with heavy news coverage (over 150 articles per year) and those with less coverage to understand how news frequency affects prediction accuracy.

An important aspect analyzes which factors matter most for predictions. This research compares sentiment signals against traditional market indicators, tests which news sources provide more reliable information, and examines whether broad market sentiment proves more useful than stock-specific news.

This research ultimately seeks to test whether any successful predictions can translate into profitable trading strategies. This includes examining trading on high-volume news coverage, using only high-quality news sources, combining sentiment signals from different market levels, and testing different strategies for heavily and lightly covered stocks.

1.3 Research Contributions

A key contribution of this research is a novel approach to combining sentiment signals from individual stocks, industries, and the broader economy. The methodology weights industry sentiment using both individual stock sentiments (70%) and direct industry news (30%), applying a similar approach at the economy level.

Additionally, this study provides a thorough comparison of modern language models for financial sentiment analysis. While large language models perform better at understanding financial texts, this advantage doesn't necessarily lead to better stock predictions. This study reveals unexpected patterns in how news coverage affects prediction quality. In some cases, stocks with less news coverage provide more reliable signals, possibly due to reduced noise in the information flow.

Through extensive testing, the study uncovers important challenges in sentiment-based prediction. Models using both market and sentiment data tend to rely too heavily on price movements. Overall market sentiment proves more useful than individual stock sentiment. Results show greater reliability for stocks with steady, quality news coverage. Established news sources don't consistently provide the most reliable sentiment signals.

The comprehensive back testing includes transaction costs, showing limited profit potential for daily sentiment-based trading strategies during 2009-2023, though some approaches focusing on less-covered stocks show modest promise.

These findings have implications for both research and practical trading. For researchers, the framework offers a new way to study information flow through markets. For traders, the results demonstrate which data sources and approaches prove most reliable, while setting realistic expectations about potential returns.

2 Literature Review

2.1 Stock Return Prediction

The idea that stock returns can be predicted has been debated for decades. Ever since Fama introduced the Efficient Market Hypothesis Fama [1970], researchers have grappled with a fundamental question: Is it even possible to consistently predict stock returns? The EMH suggests that stock prices already reflect all available information, making prediction essentially impossible. Later work by Fama and French [1992, 1993] introduced their famous three-factor model, showing that certain market anomalies seemed to persist despite market efficiency.

Traditional technical analysis, which tries to find patterns in historical price data, has largely failed to deliver consistent results. Brock et al. [1992] initially found promising results with moving averages and trading range breaks, but Sullivan et al. [1999] showed these patterns disappeared once they accounted for data mining bias. Most technical indicators work in some periods but break down in others, supporting the adaptive market hypothesis proposed by Lo [2004].

Machine learning approaches initially seemed more promising. Early studies by Kimoto et al. [1990] using neural networks and Tay and Cao [2001] with support vector machines reported impressive results, but most of these didn't hold up in real-world trading. As shown by Bailey et al. [2014], many of these early successes suffered from backtest overfitting.

Time series models like ARIMA have their own limitations. While Campbell and Thompson [2008] showed they can capture some short-term price movements, these models struggle with sudden changes in market conditions. Cont [2001] documented that financial returns exhibit properties like heavy tails and volatility clustering that simple time series models can't capture.

The rise of deep learning, particularly LSTM networks Hochreiter and Schmidhuber [1997], brought renewed optimism. Fischer and Krauss [2018] showed promising results using LSTM for daily stock predictions. However, as Zhang et al. [2019] demonstrated, even sophisticated deep-learning models struggle with the fundamental unpredictability of daily stock returns.

2.2 Sentiment Analysis in Finance

The development of sentiment analysis in finance reflects overall progress in natural language processing. Early approaches used simple word counts with dictionaries - from the Harvard IV-4 dictionary Stone et al. [1966] to the finance-specific Loughran-McDonald dictionary Loughran and McDonald [2011]. Tetlock [2007] showed that even these basic approaches could predict market returns.

Machine learning methods brought improvements. Antweiler and Frank [2004] used Naive Bayes on internet message boards, while Pang et al. [2002] pioneered Support Vector Machines for sentiment classification. However, as Das and Chen [2007] noted, these models struggled with financial jargon and context-dependent meanings.

Word embeddings represented a major step forward. Mikolov et al. [2013] introduced Word2Vec, while Le and Mikolov [2014] developed Doc2Vec. In finance, Ding et al. [2015] showed how word embeddings could capture complex relationships between financial terms and market movements.

The real breakthrough came with transformer models. BERT Devlin et al. [2019] and its financial variants like FinBERT Araci [2019] and AlphaBERT Huang et al. [2022] showed impressive results in understanding complex market commentary. Yang et al. [2020] demonstrated how these models could capture subtle nuances in earnings calls and financial news.

2.3 Machine Learning in Financial Markets

Using machine learning in financial markets presents specific challenges. For feature engineering, Gu et al. [2020] showed the importance of combining traditional market data with alternative data sources. Chen et al. [2019] demonstrated how news sentiment could enhance traditional factors.

Model selection is another critical challenge. While Chinco et al. [2019] showed the benefits of complex non-linear models, López de Prado [2018] warned about their tendency to overfit. Green et al. [2013] documented over 300 factors claimed to predict returns, but Harvey et al. [2016] found most couldn't survive rigorous statistical testing. The overfitting problem is particularly acute in finance. As Bailey et al. [2014] showed, the vast amount of financial data makes it easy to find spurious patterns. This aligns with Wolpert's "no free lunch" theorem Wolpert and Macready [1997] - more complex models aren't necessarily better at predicting returns.

2.4 Multi-level Market Analysis

Markets operate at multiple levels simultaneously. Roll [1988] showed that individual stock movements are influenced by industry and market factors. Cohen and Frazzini [2008] documented how information flows between economically linked firms can create predictable patterns.

Industry-wide effects can be powerful predictors. Hong et al. [2007] showed how earnings surprises in one firm affect other firms in the same industry. Kelly and Pruitt [2013] developed a hierarchical framework for combining signals across different market levels.

Market-wide sentiment also plays a crucial role. Baker and Wurgler [2006] constructed a market sentiment index that predicts returns. During periods of high market stress, King [1966] and later studies showed correlations between stocks tend to increase.

Cross-asset relationships add another layer of complexity. Andersen et al. [2007] documented how macroeconomic news affects multiple markets simultaneously. The hierarchical approach builds on these insights while acknowledging the dynamic nature of these relationships documented by Adrian and Brunnermeier [2016].

3 Data and Methodology

3.1 Data Sources

3.1.1 News Data

The news data used in this study comes from three main sources, each with its own characteristics and coverage patterns. The core dataset is FNSPID Dong et al. [2024], which provides 15.7 million financial news records covering S&P 500 companies from 1999 to 2023. This dataset is particularly valuable because its news items are already time-aligned with stock market events.

FNSPID is supplemented with the Edaz financial news dataset, containing over 11 million Yahoo Finance articles spanning from pre-2000 to 2023. The third source is Philippe Remy's dataset of more than 500,000 Bloomberg articles and Reuters headlines from 2006 to 2013, which offers high-quality coverage from traditional financial news providers.

The news sources in the dataset display interesting patterns in their coverage and style. Reuters and Bloomberg maintain their reputation for neutral, fact-based reporting. In contrast, sources like Nasdaq, Benzinga, and Zacks tend toward more attention-grabbing headlines. Rather than discard the potentially "clickbait" sources, they are retained as they might capture retail investor attention - a factor that could influence short-term price movements.

Looking at the distribution of news coverage across companies, a highly skewed pattern is found. In a log-log plot of articles per year, the vast majority of firms receive less than 100 articles annually, as shown in Figure 1. This observation led to the development of separate modeling approaches for high and low-coverage stocks, as the information dynamics likely differ between frequently and rarely covered companies.

3.1.2 Market Data

For market data, Yahoo Finance's API is used to collect daily stock prices. This includes open, high, low, and close prices, and trading volume for all U.S. stocks currently trading. A significant limitation of the dataset is survivorship bias. While data was collected for all stocks currently trading (over 5,000), only about 2,500 of them have data going back to 2009, the start of the study period. This means the dataset only includes companies that have "survived" from 2009 to today, excluding companies that went bankrupt, were delisted, or were acquired during this period as shown in Figure 2.

This survivorship bias likely makes the results appear more optimistic than they would be in real-world implementation. Companies that survived for 14+ years tend to be more stable and successful than the average company, potentially making their price movements more predictable. In practice, a trader in 2009 wouldn't have known which companies would survive, making the actual trading problem considerably more challenging.

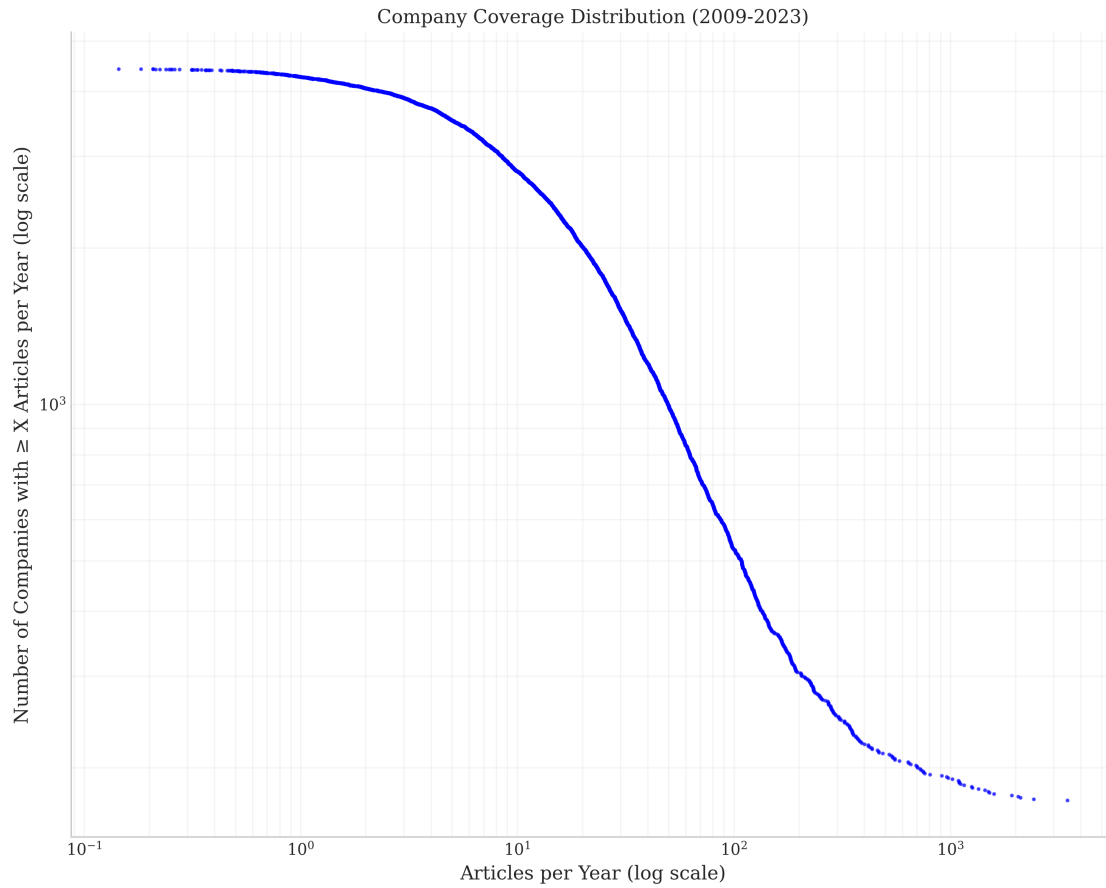


Figure 1: Log-Log Company Coverage Distribution.

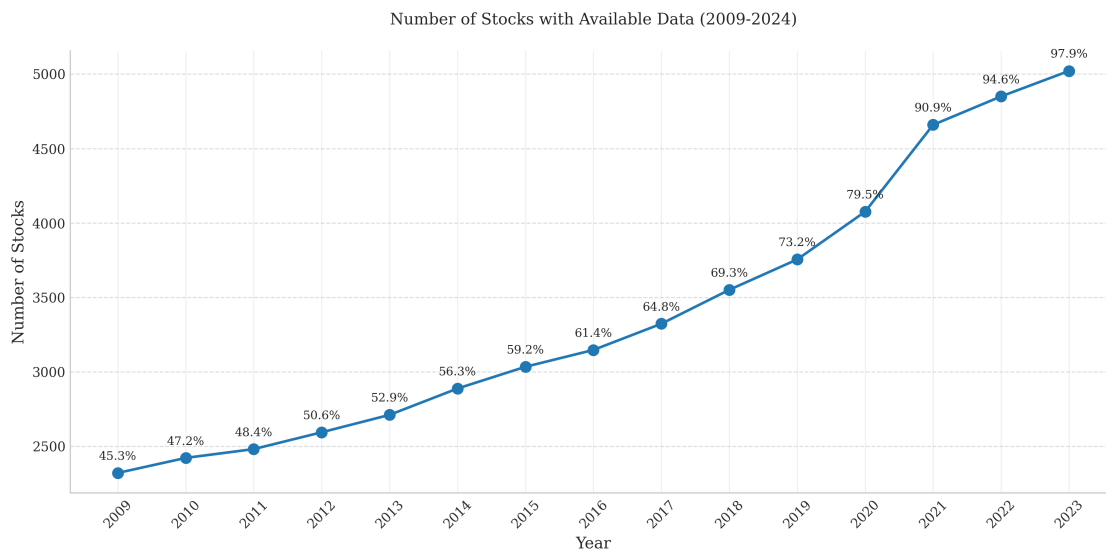


Figure 2: Number and proportion of stocks with available price data.

Adjusted close prices are used to account for corporate actions like splits and dividends. The adjustments follow Yahoo Finance's methodology, which recalculates historical prices to reflect these events. This ensures price series are

comparable across time, though it does mean the historical prices in the dataset differ from the actual quoted prices on those dates.

3.2 NLP Model Selection and Benchmarking

3.2.1 Model Comparison

The evaluation encompassed four state-of-the-art language models for financial sentiment analysis:

1. RoBERTa Large: A robustly optimized BERT model demonstrating strong performance on general NLP tasks.
2. FinBERT: A domain-adapted model specifically designed for financial text analysis.
3. LLaMA 3 8B: A newer architecture offering competitive performance with reduced parameter count.
4. Instruction-pretrained LLaMA 3 8B: An enhanced variant leveraging recent advances in instruction tuning Cheng et al. [2024].

3.2.2 Benchmark Results

The models were evaluated on two standard financial sentiment datasets: FPB (Financial Phrase Bank) and FiQA (Financial Question Answering). The benchmark testing revealed:

- FinBERT demonstrated robust performance (84.2% accuracy on FPB, 74.7% on FiQA) despite its relative age
- LLaMA 3 8B achieved the highest accuracy scores (97.0% on FPB, 84.4% on FiQA)
- Instruction-pre trained LLaMA 3 8B showed comparable results (96.4% on FPB, 76.5% on FiQA)

While LLaMA 3 8B demonstrated marginally superior performance metrics, the instruction-pretrained variant was selected for subsequent analysis. The model's domain adaptation through instruction tuning suggests enhanced robustness for real-world financial texts, even though this advantage is not fully captured in the benchmark scores. Table 1 shows the FPB & FIQA finetuning results.

Table 1: FPB & FIQA finetuning results

	FPB Accuracy	FPB F1 Score	FIQA Accuracy	FIQA F1 Score
Roberta Large	0.285	0.131	0.651	0.518
FinBert	0.842	0.844	0.747	0.762
Llama 3 8B	0.9701	0.9699	0.844	0.858
Finance Instruct PT	0.9643	0.9642	0.765	0.796

3.3 Data Processing Pipeline

3.3.1 Text Processing

The text processing pipeline begins with timestamp conversion to EST to ensure alignment with U.S. market hours. This standardization is crucial as after-hours news must be appropriately attributed to the next trading day.

Entity detection employs a simplified approach based on company name matching. Rather than implementing complex Named Entity Recognition (NER) systems, the methodology utilizes a dictionary of shortened company names (e.g., "Coca" for "The Coca-Cola Company", "Rivian" for "Rivian Automotive"). While this approach may miss certain mentions, it demonstrates greater resistance to false positives compared to NER systems that might, for example, misidentify "Apple & Google" as a single entity.

Industry references are identified through a curated list of industry keywords (e.g., "Oil market", "Automotive industry"), while economic topics are detected via terms such as "FED" or "US Dollar". The system allows multiple entity tags per article.

Article summarization implements the LexRank algorithm to extract the five most relevant sentences. The selection of extractive over abstractive summarization reflects the importance of preserving original phrasing for accurate sentiment analysis in financial contexts. The algorithm identifies representative content based on graph centrality measures.

3.3.2 Market Data Processing

A central challenge in market data processing involves the alignment of price data with news flow. The analysis employs close-to-close returns as the target variable, calculated as:

$$R_{t+1} = \frac{P_{t+1}}{P_t} - 1$$

where P_t represents the adjusted closing price on day t .

News articles published after 4:30 PM EST are attributed to the subsequent trading day, acknowledging that their impact cannot be reflected in the current day's closing price. This cutoff time, positioned slightly after market close, accounts for end-of-day news that may still pertain to the current trading session.

4 Model Development

4.1 Feature Sets

4.1.1 Sentiment Features

The sentiment feature engineering methodology adapts to varying levels of news coverage, with distinct approaches for high-coverage stocks (>150 articles annually) and low-coverage stocks. Features are developed at three distinct levels - stock, industry, and economy - to capture the hierarchical nature of market information flow.

At the stock level, daily sentiment aggregates include:

- Mean sentiment score
- Maximum and minimum sentiment scores
- Sentiment dispersion (max-min spread)
- Article count and sentiment-weighted article count
- Time since last news mention

For high-coverage stocks, rolling sentiment metrics capture short and medium-term movements across 5, 21, and 42-day windows. These include sentiment momentum (changes in rolling averages), volatility measures tracking sentiment stability, and source agreement metrics measuring consistency across news providers. Empirical analysis suggests these features often precede significant price movements in frequently covered stocks.

Low-coverage stocks require a different approach emphasizing stability over responsiveness. The methodology employs longer-term sentiment averages spanning 21 to 84 days, complemented by measures of sentiment consistency and coverage intensity. Given the sporadic nature of news for these securities, particular attention is paid to inter-article gaps and source agreement metrics when multiple sources are available.

For industry aggregation, a weighted approach combines individual stock sentiments with direct industry news:

$$S_{\text{ind}} = 0.7 \cdot \sum_{i \in \text{ind}} w_i S_i + 0.3 \cdot S_{\text{ind, direct}}$$

where w_i represents the normalized trading volume weight of stock i within its industry, S_i is the stock's sentiment score, and $S_{\text{ind, direct}}$ captures sentiment from industry-wide news articles.

Economy-wide sentiment follows a similar pattern:

$$S_{\text{econ}} = 0.7 \cdot \sum_j v_j S_j + 0.3 \cdot S_{\text{econ, direct}}$$

where v_j represents industry weights and $S_{\text{econ, direct}}$ comes from broad economic news.

The 70-30 split between aggregated lower-level sentiment and direct sentiment reflects the assumption that individual component sentiments provide more granular information than broad coverage. Testing suggests the exact ratio isn't critical as long as both components are represented. Detailed feature definitions and implementation specifics are provided in Appendix A.

4.1.2 Market Features

While sentiment analysis forms the core of this research, market data provides essential context for sentiment signals. This integration presents a significant methodological challenge, as models demonstrate a tendency to over fit on price-based signals when available, potentially obscuring sentiment-based predictive power. The framework incorporates standard market metrics including multi-horizon returns, normalized volume indicators, and technical signals. Significant value derives from interaction features connecting sentiment and market data, particularly:

- Sentiment-volume interactions capturing attention effects
- Sentiment-volatility interactions measuring uncertainty impact
- Cross-level sentiment divergence (e.g., stock vs. industry or stock vs economy sentiment difference)
- Price momentum and sentiment trend alignments

Industry and market-wide features follow similar principles, utilizing ETF data for industry groups and S&P 500 data for broad market metrics. This approach maintains methodological consistency across analysis levels while preserving the unique characteristics of each market segment. Detailed feature definitions and implementation specifics are provided in Appendix A.

4.2 Model Architecture

4.2.1 Coverage-Based Approach

Initial development focused on separate models for high and low-coverage stocks, based on distributional analysis of news frequency. The high-coverage model targeted stocks with more than 150 articles annually, while the low-coverage model addressed stocks with 5-50 annual articles. Stocks with 51 to 146 annual articles would be predicted using a weighted average of both high and low-coverage models. The hypothesis suggested these groups would benefit from differentiated modeling approaches due to their distinct information environments.

The high-coverage implementation utilized a CatBoost model with deep trees (depth 10) and extensive feature interactions, justified by data abundance. Despite the implementation of gradient boosting with feature dropouts and comprehensive cross-validation, persistent overfitting emerged - evidenced by exceptional training performance but poor validation results.

The low-coverage implementation adopted a more conservative approach with reduced tree depth (5), limited feature interactions, and increased reliance on industry-level features to compensate for sparse stock-specific data. While this model demonstrated greater stability, performance advantages over a unified approach proved insignificant.

Extensive testing over a three-month period led to the adoption of a single, conservative model for all stocks. The final CatBoost configuration employs 200 iterations with 0.005 learning rate, depth 7, and L2 regularization weight of 10. Aggressive sub sampling (0.6 for instances, 0.5 for features) combined with moderate randomization through the bagging temperature of 1 provides robust performance across varying coverage levels.

4.2.2 Model Training

The training methodology evolved through multiple iterations, ultimately settling on a monthly retraining schedule with a two-year look back period. The selection of monthly retraining frequency reflects several key considerations:

First, market microstructure analysis indicates that relationship patterns between sentiment and returns typically evolve over weeks to months rather than days. Daily retraining, while theoretically more responsive, introduces excessive noise and computational overhead without commensurate performance improvements. Second, monthly retraining aligns with typical institutional investment cycles, particularly regarding factor model updates and risk parameter estimation. This synchronization facilitates integration with existing quantitative trading frameworks.

Third, empirical testing across various retraining frequencies (daily, weekly, monthly, quarterly) reveals monthly updates provide optimal balance between model stability and adaptiveness to changing market conditions. More frequent updates lead to parameter instability, while longer periods risk missing evolving market dynamics.

The two-year lookback window, selected after testing periods from 1 to 3 years, ensures sufficient data for stable parameter estimation while maintaining responsiveness to market evolution. Each training cycle utilizes approximately 504 trading days of historical data for predicting the subsequent 21 trading days.

The validation framework employs a rolling window methodology, training on two years of data, validating on the subsequent month, then rolling forward. This approach better replicates actual trading conditions compared to traditional

cross-validation methods, which can introduce look-ahead bias in time series data. A minimum requirement of 252 trading days per stock ensures stable parameter estimation.

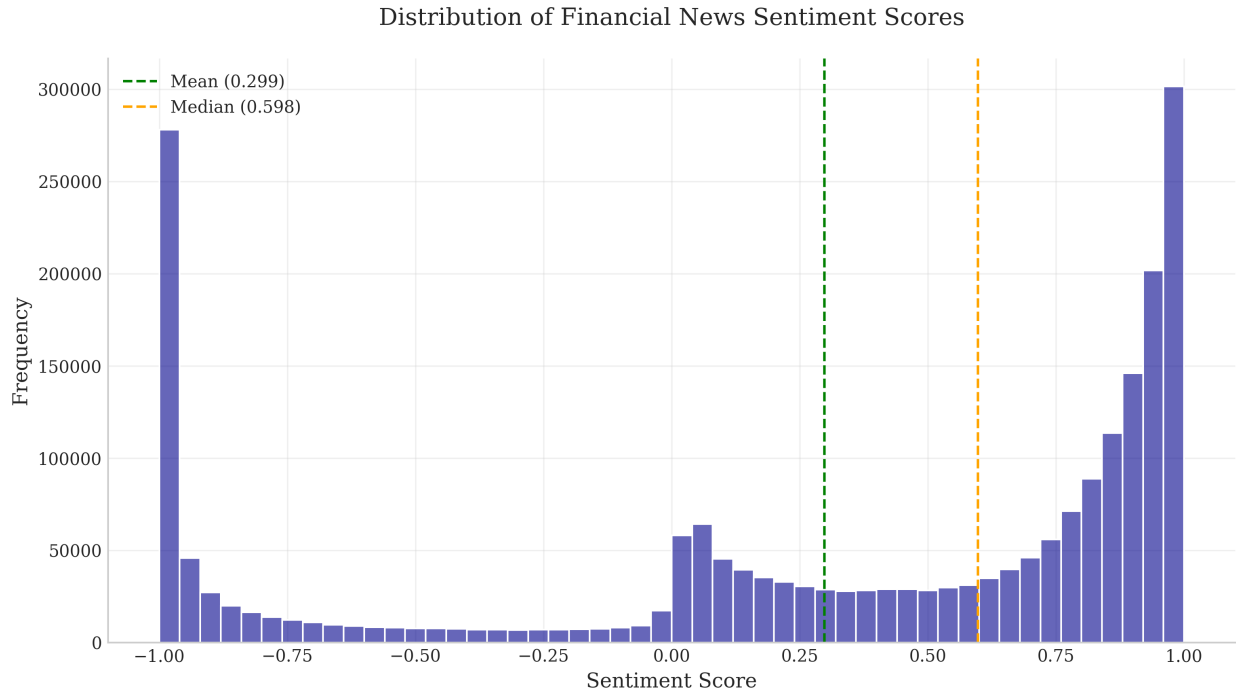
Model monitoring encompasses R-squared values and directional accuracy metrics. Regular observation of feature importance stability and prediction distributions helps identify potential model degradation before it impacts performance.

5 Results and Analysis

5.1 Sentiment Analysis

5.1.1 Sentiment Analysis Signal

Analyzing financial news from 2009 to 2023 shows clear patterns in both sentiment distribution and the reliability of sources. The sentiment score distribution exhibits notable clustering around -1, 0, and 1, consistent with the model's training objectives. Both mean and media sentiment scores demonstrate a significant positive bias, potentially reflecting the inherently optimistic nature of financial news discourse. This systematic positive bias presents particular challenges for sentiment-based trading strategies, as it necessitates careful calibration of sentiment thresholds and suggests that relative sentiment changes may be more informative than absolute values, illustrated in Figure 3.



Metric	Value
Mean	0.299
Median	0.598
Std Dev	0.713
Sample Size	2,201,934

Figure 3: Distribution of Financial News Sentiment Scores.

Comparing sentiment assessments among major news sources reveals significant differences in how they interpret sentiment. For instance, examining Apple (AAPL) coverage demonstrates that daily sentiment agreement between

primary sources such as Reuters, Bloomberg, and Benzinga ranges from 47% to 60%, marginally exceeding random chance as shown in Figure 4. This limited consensus has profound implications for sentiment-based trading strategies. First, it suggests that aggregating sentiment across multiple sources may not provide the noise reduction benefits typically expected from averaging. Second, it indicates that different news sources might be capturing distinct aspects of market sentiment, potentially warranting source-specific modelling approaches.

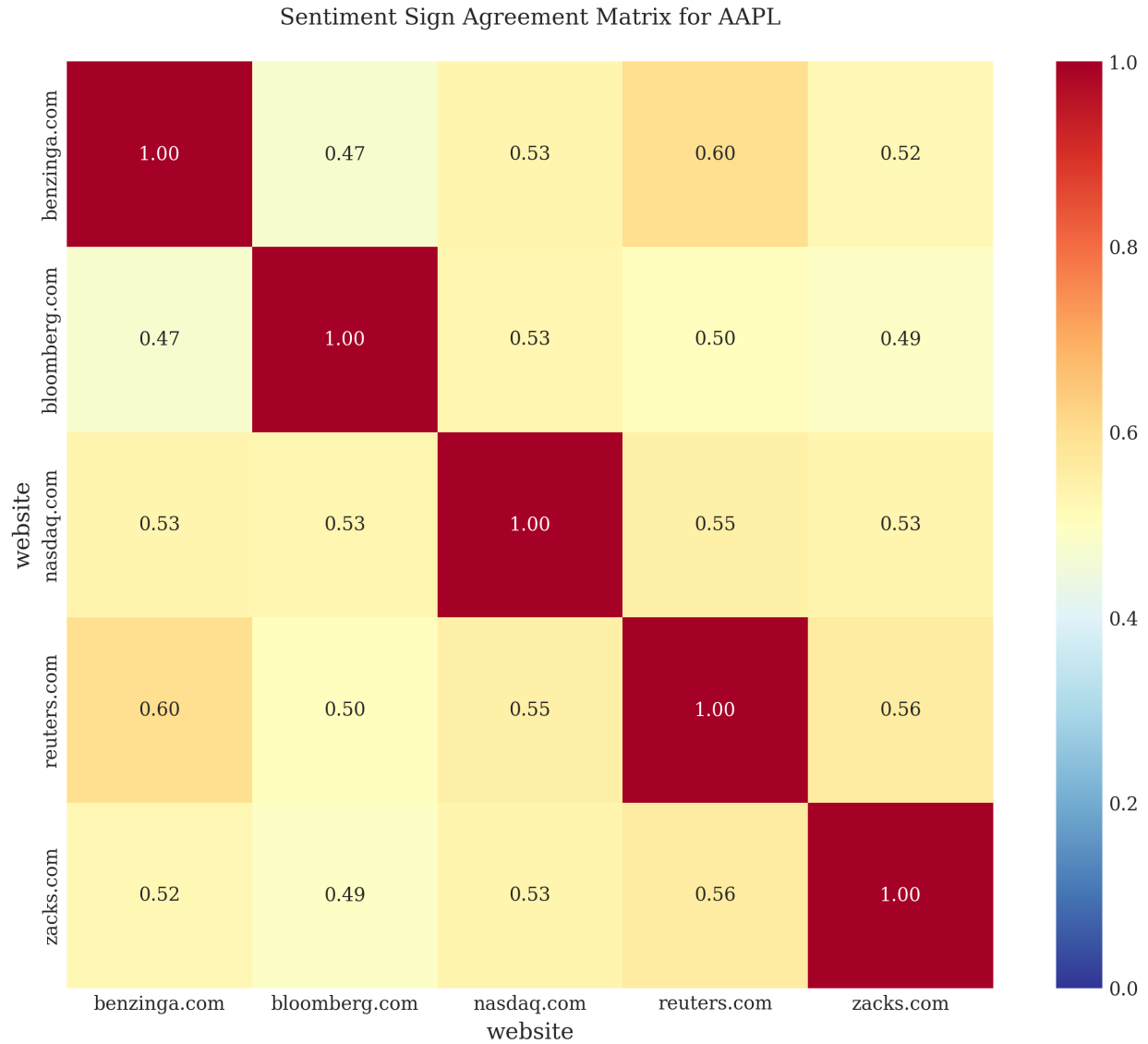


Figure 4: Sentiment Sign Agreement Matrix for AAPL.

Over time, sentiment patterns vary systematically across different news sources. Bloomberg consistently exhibits the most negative sentiment bias, while Reuters demonstrates a notable transition from predominantly positive to predominantly negative sentiment over the study period. The observed shift in Reuters' sentiment profile is particularly intriguing, potentially reflecting changes in editorial policy, market conditions, or broader shifts in financial journalism. Most surprisingly, sources typically categorized as retail-oriented (Zacks, Nasdaq, and Benzinga) maintain the highest temporal stability in sentiment profiles. This stability proves crucial for predictive modeling, as it provides more reliable signals for machine learning algorithms, potentially explaining these sources' enhanced predictive utility despite their perceived lower journalistic prestige. Figure 5 shows yearly average sentiment by news source.

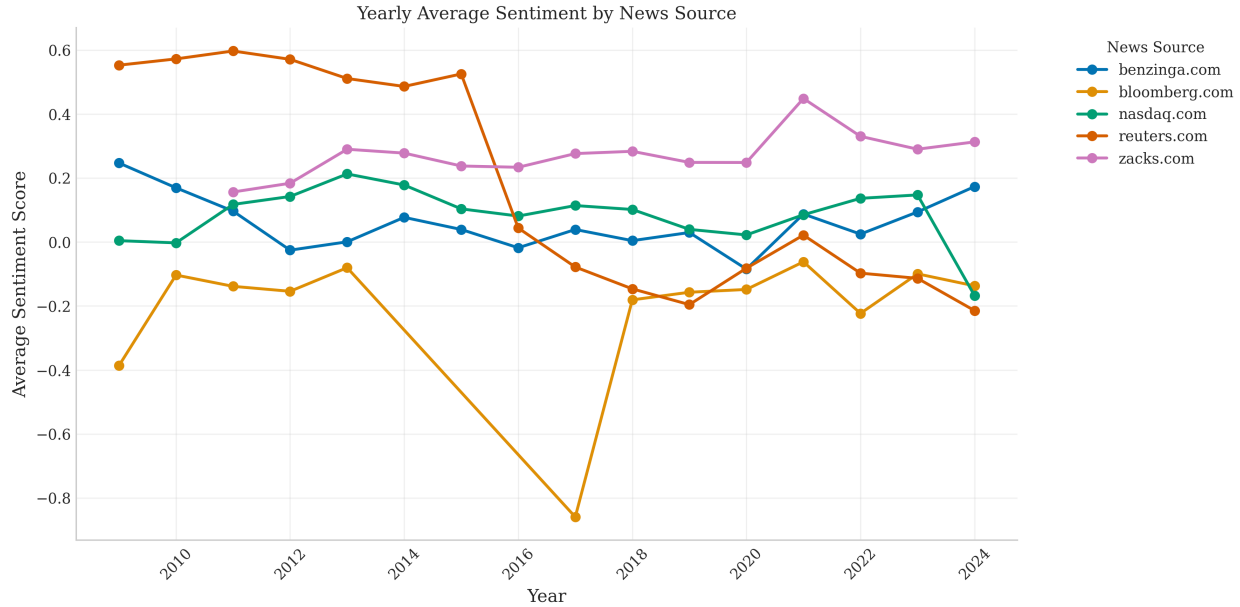


Figure 5: Yearly Average Sentiment by News source.

5.1.2 Sentiment-Stock Return Agreement

Exploring the link between sentiment and stock returns highlights significant challenges in using sentiment alone for predictions. Analysis of stocks with minimum 50 observations shows sentiment-return sign agreement ratios clustering around 0.5, approximating random distribution. This finding challenges the common assumption that sentiment directly leads price movements at daily frequencies. The lack of significant predictive power in raw sentiment signals suggests that more sophisticated approaches, incorporating multiple time horizons and market contexts, maybe necessary for effective sentiment-based trading.

Examining changes over time in sentiment-return relationships suggests evolving market efficiency. While mean agreement rates remain consistently proximate to 0.5, the standard deviation of agreement rates demonstrates a substantial decline from 0.2 in 2009 to 0.065 in 2023. This reduction in volatility represents a fundamental shift in market microstructure. The decreasing variance in sentiment-return relationships likely reflects the maturation of sentiment-based trading strategies and increasing market efficiency in processing new information. This trend has crucial implications for strategy development, suggesting that simple sentiment-based approaches that may have worked in the past are unlikely to maintain effectiveness in current market conditions. Figure 7 illustrates time Series of Average Sentiment-Return Sign Agreement.

5.2 Prediction Performance

5.2.1 Model Performance Analysis

Testing models from 2015 to 2022, using two-year training windows, shows distinct performance differences among various modeling methods. While linear models show better overall results with an average test R-squared of -0.013 compared to CatBoost's -0.311, this appears to reflect underfitting in the linear model. CatBoost demonstrates higher potential with a maximum test R-squared of 0.005 and superior directional accuracy of 0.548 versus 0.526 for the linear model, as shown in Figure 6. Given CatBoost's ability to capture more complex patterns when properly parametrized, I chose to use it for the remainder of the study despite its great versatility in performance. The final configuration employs conservative hyper parameters to balance learning capacity with stability. This pattern merits careful consideration in model selection. While CatBoost's superior maximum performance might seem attractive, the greater stability and coefficient interpretability of linear models suggests they may be more suitable for practical implementation, particularly given the noise inherent in sentiment data. Table 2 shows the comparison between CatBoost vs Linear model.

Feature importance analysis provides insights about the models' learning behaviour. When using all available features, the CatBoost model shows clear signs of overfitting on price-based features, with a training R-squared of 0.23 dropping dramatically to -0.008 in testing. In contrast, using only sentiment features produces more stable results with training

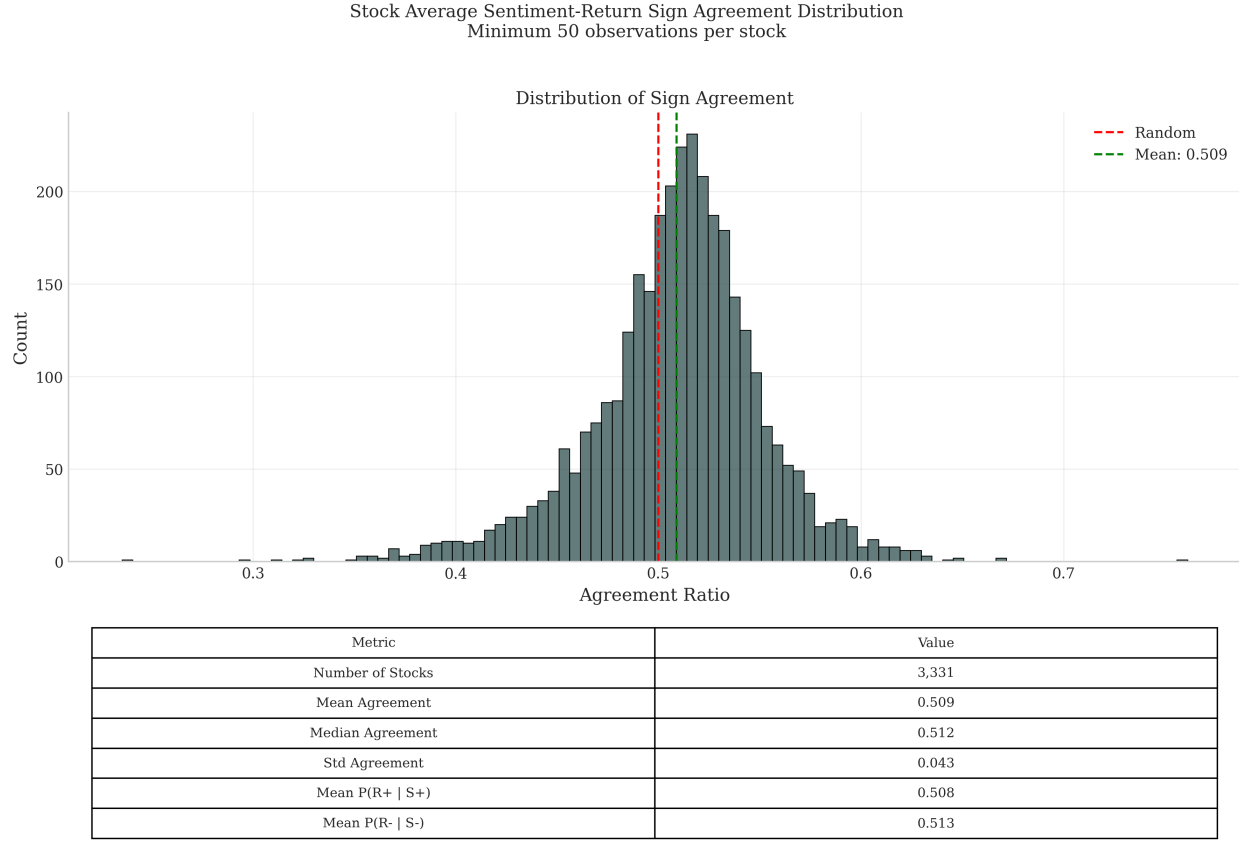


Figure 6: Stock Average Sentiment-Return Sign Agreement Distribution

Table 2: CatBoost vs Linear model performance

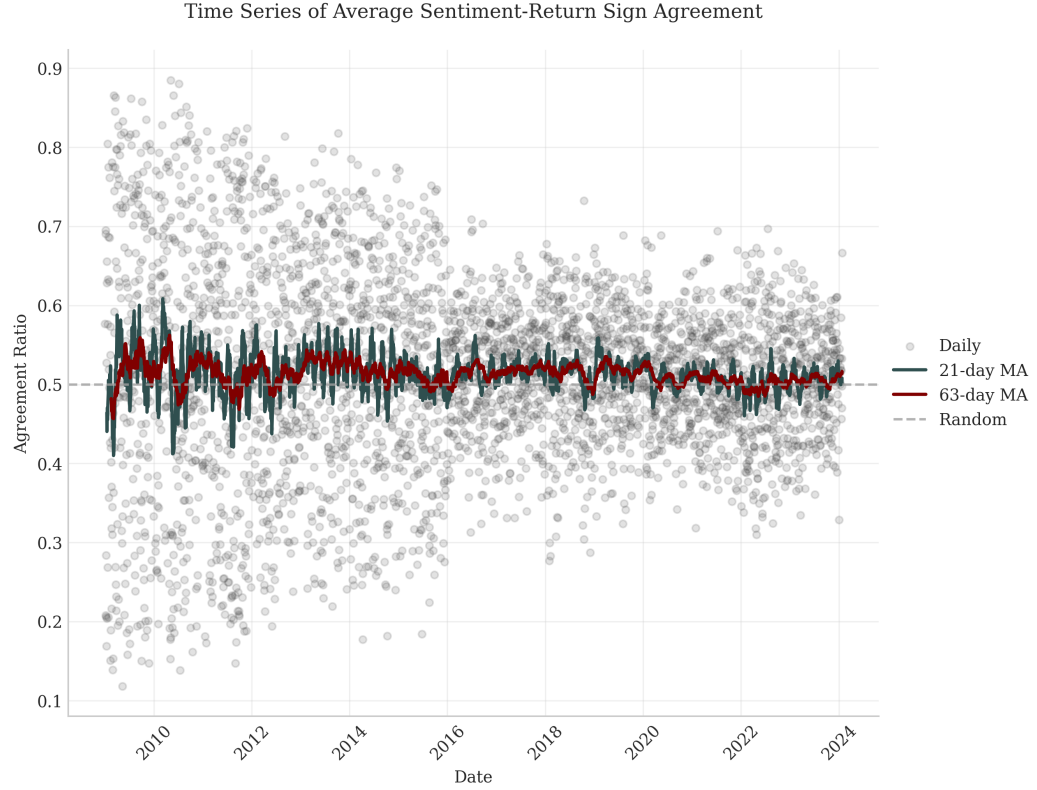
Model	Test R2Average	Test R2Max	Test R2Min	Test Directional AccuracyAverage
CatBoost	-0.311	0.005	-2.467	0.548
Linear	-0.013	-0.001	-0.044	0.526

R-squared of 0.02 and test R-squared of -0.004, suggesting better generalization despite lower absolute performance. This pattern indicates that while price features contain predictive information, the model's tendency to overfit on these signals can obscure the more subtle but potentially more stable sentiment relationships. The dominance of price-based features over sentiment indicators suggests that market prices remain the most informative predictors of future returns, consistent with efficient market theory. Second, market-level signals consistently outperform stock-specific metrics in both price and sentiment domains. This hierarchy of feature importance challenges the common practice of focusing on stock-specific sentiment analysis and suggests that broader market context provides more reliable predictive signals. Table 3 lists the performance comparison between All Features and Sentiments Only.

Table 3: All Features vs Sentiments features CatBoost performance

Model	Train R2 average	Test R2 Average	Test R2 Max	Test R2 Min	Test Directional Accuracy Average
All Features	0.23	-0.008	0.002	-0.031	0.501
Sentiments Only	0.02	-0.004	0.004	-0.015	0.517

Coverage-based analysis yields surprising results, with low-coverage stocks (5-50 annual articles) demonstrating superior predictive performance compared to high-coverage stocks(>168 annual articles). This finding has significant



Metric	Full Period	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Mean	0.513	0.521	0.519	0.505	0.513	0.531	0.515	0.507	0.516	0.519	0.513	0.522	0.506	0.507	0.497	0.507	0.514
Std	0.122	0.201	0.182	0.189	0.143	0.135	0.137	0.123	0.072	0.067	0.086	0.070	0.063	0.069	0.084	0.065	0.066

Figure 7: Time Series of Average Sentiment-Return Sign Agreement

implications for strategy development. The superior predictability of low-coverage stocks suggests potential market inefficiencies in sentiment processing for less frequently covered securities, possibly due to slower information diffusion or less algorithmic trading activity in these stocks. Moreover, this performance gap might be partially explained by the highly skewed distribution of news coverage across stocks -while only a small subset of companies receive extensive media attention (>168 articles annually), the majority of stocks fall into the low-coverage category. This imbalance means the high-coverage model must learn from a significantly smaller set of training examples, potentially limiting its ability to generalize effectively compared to the low-coverage model, which benefits from a larger and more diverse training set. Table 4 shows the comparison of high vs low coverage stocks performance.

Table 4: High Performance on high vs low coverage stocks

	Test R2 Average	Test R2 Max	Test R2 Min	Test Directional Accuracy Average
High Coverage	-0.006	0.001	-0.029	0.502
Low Coverage	-0.003	0.001	-0.009	0.51

Comparative analysis of news sources gave unexpected predictive superiority of retail-oriented sources (Benzinga, Zacks, Nasdaq) over traditional financial news providers. This performance differential correlates with observed temporal stability patterns in sentiment reporting and challenges conventional wisdom about news source quality. The superior predictive power of retail-oriented sources might reflect their focus on market-moving information or their ability to capture retail investor sentiment more effectively. Table 5 lists the performance using Reuters vs Other News sources.

Table 5: Performance using Reuters vs Other News Sources

	Test R2 Average	Test R2 Max	Test R2 Min	Test Directional Accuracy Average
Reuters	-0.013	0.002	-0.031	0.505
Others	-0.004	0.002	-0.013	0.501

Predictive effectiveness fluctuates noticeably over time throughout the study period. The years 2015 and 2018 stand out as particularly challenging, with R-squared values dropping to negative value and directional accuracy barely exceeding random chance. These periods coincide with significant market volatility and regime shifts - 2015 saw the Chinese stock market crash and fears of Fed tightening, while 2018 was marked by trade war tensions and a sharp market correction in Q4. In contrast, 2019 consistently emerges as the strongest period for prediction accuracy, with R-squared values reaching 0.004 and directional accuracy exceeding 53%. This superior performance might be attributed to the more stable market environment and clearer sentiment trends during this period. Additional pockets of strong performance appear in 2017 and 2020, though the latter's success is somewhat surprising given the market turbulence during the COVID-19 pandemic. This temporal variation in performance reminds the importance of market regime consideration in sentiment-based strategies and suggests that the effectiveness of sentiment signals may depend heavily on broader market conditions.

5.2.2 Strategy Performance

Backtest results analysis indicates superior performance from a specialized approach focusing on sentiment features, low-coverage stocks, and implementing a 0.1 correlation threshold in training sets. Performance metrics demonstrate moderate but consistent results: average Sharpe ratio of 0.609, mean total return of 8.7%, maximum return of 24.4%, and maximum draw down of -29.5%. These results warrant careful interpretation. While not exceeding market benchmarks, they demonstrate statistically significant predictive capacity in filtered sentiment signals. The modest performance metrics suggest sentiment analysis might be more valuable as a component of a broader strategy rather than as a standalone approach.

Implementation challenges emerge primarily in transaction cost impact and position sizing optimisation. High portfolio turnover significantly affects net returns, while position sizing faces dual constraints: market impact in less liquid securities and minimum scale requirements for cost efficiency. These practical limitations highlight the gap between theoretical predictability and real-world profitability in sentiment-based trading strategies.

5.3 Sentiment Reversal Analysis

5.3.1 Event Analysis

Detailed examination of significant market events in 2021-2023 reveals a consistent pattern of divergence between sentiment signals and price movements, particularly around major corporate announcements. The analysis demonstrates that positive sentiment often fails to translate into corresponding price appreciation, challenging conventional assumptions about sentiment-based prediction.

Microsoft's Q3 2021 earnings report (April 28, 2021) provides a particularly instructive example. Despite positive sentiment surrounding strong performance in cloud computing and software services, the stock price reacted negatively. This divergence suggests that market participants may have already priced in strong results or harbored concerns about future growth rates and competitive pressures that weren't captured in the sentiment analysis.

Similarly, Apple's Q1 2021 earnings report (January 27, 2021) exhibited this disconnect. While the report generated positive sentiment due to growth in iPhone sales, wearables, and services, the stock price declined. This pattern indicates that even when fundamental performance aligns with positive sentiment, other factors such as valuation levels, supply chain constraints, or concerns about sustaining growth rates can dominate price action.

5.3.2 Pattern Recognition

Analysis across multiple events reveals several recurring patterns:

1. **Sentiment-Expectation Mismatch:** Johnson & Johnson's Q3 2023 earnings report exemplifies how positive sentiment around robust sales performance can be overshadowed by broader market concerns. Despite favorable sentiment scores, the stock declined due to underlying worries about regulatory challenges, cost pressures, and future product pipeline uncertainties.
2. **Valuation-Sentiment Disconnect:** The case of Coca-Cola's BodyArmor acquisition (November 1, 2021) demonstrates how positive sentiment around strategic moves doesn't necessarily translate to price appreciation. While the market viewed the expansion into sports drinks favorably, the neutral price reaction suggests investors saw it as an incremental rather than transformative development.
3. **Market Sophistication:** A common thread across these events is the market's ability to look beyond headline sentiment. For instance, Microsoft's case shows how investors may focus on subtle guidance cues or competitive dynamics rather than reported results. This sophistication in market response makes simple sentiment-based trading strategies increasingly challenging.

The analysis suggests that sentiment signals, while informative, often fail to capture the full complexity of market positioning and investor expectations. The recurring pattern of positive sentiment followed by neutral or negative price action indicates that sophisticated market participants may be positioning themselves contrary to obvious sentiment signals, particularly around significant corporate events.

These findings reinforce the broader conclusion that successful trading strategies need to consider sentiment as just one component of a more comprehensive analysis framework, rather than relying on it as a primary signal for price prediction.

5.4 Trading Strategy Implementation

5.4.1 Portfolio Construction

Portfolio optimization employs a systematic approach solving for optimal position weights while respecting practical trading constraints. The base capacity parameter of one billion USD serves as a global constraint on strategy size, while individual position limits prevent concentration in any single name. The optimization framework incorporates both expected returns derived from the sentiment model and trading constraints based on market liquidity.

The position sizing algorithm explicitly considers average daily trading volume when determining maximum allowable positions. By limiting individual positions to no more than 10% of typical daily volume, the strategy maintains the ability to adjust positions without creating excessive market impact. This volume-based constraint proves particularly important for low-coverage stocks where sentiment signals may be stronger but liquidity often constrains implementable position sizes.

Risk management within the portfolio construction process centers on a risk aversion parameter of 1.0 in the optimization objective function. This parameter balances the pursuit of sentiment-indicated returns against portfolio volatility, resulting in more diversified position that would be obtained from pure return maximization.

5.4.2 Execution Framework

The backtesting engine implements daily portfolio rebalancing while accounting for practical trading constraints. Each day, the system generates updated sentiment scores and target positions, then simulates trade execution considering both transaction costs and market impact. The implementation assumes a conservative 10 basis point trading cost for both entry and exit trades, reflecting typical institutional execution costs.

A critical aspect of the execution framework involves the treatment of after-hours news flow. The system considers 4:30 PM EST as a cut-off time for same-day trading decisions. Any sentiment signals generated from news arriving after this cutoff influence trading decisions for the next business day. This timing adjustment proves especially important during earnings seasons when many companies release information after market hours.

6 Discussion

6.1 Key Findings

This analysis reveals several fundamental challenges in using sentiment analysis for next-day stock return prediction. Despite advances in NLP models and the development of sophisticated multi-level sentiment aggregation techniques, the relationship between news sentiment and short-term returns remains remarkably difficult to exploit consistently.

One of the most notable findings is the dominance of market-level sentiment signals over stock-specific indicators. When given both types of information, the models consistently assigned higher importance to broader market sentiment features. This suggests that individual stock sentiment might be too noisy for reliable prediction, while market-wide sentiment captures more stable and actionable signals. The pattern holds true for both price-based and sentiment-based features, pointing to a hierarchical structure in market information processing that favors top-down analysis.

Another surprising discovery was the superior predictive performance in low-coverage stocks compared to frequently covered securities. The initial hypothesis that more news coverage would lead to better predictions proved incorrect.

Instead, stocks with sparse but consistent news coverage often generated more reliable signals. This counterintuitive result suggests that the quality and timing of news might matter more than quantity, and that excessive news coverage could actually introduce more noise than signal into the prediction process.

The source reliability analysis yielded unexpected insights. Traditional financial news sources like Reuters and Bloomberg, despite their reputation for quality journalism, didn't provide consistently better predictive signals than retail-oriented sources like Benzinga or Zacks. In fact, the more stable sentiment profiles of retail-oriented sources often translated into more reliable prediction inputs. This finding challenges conventional wisdom about news source hierarchy and suggests that temporal stability in sentiment reporting might be more valuable for prediction than perceived journalistic quality.

Perhaps most importantly, the research found that even with advanced NLP models and careful feature engineering, the predictive power of sentiment analysis remains modest at daily horizons. The best configuration achieved a Sharpe ratio of 0.609 and an average total return of 8.7%, figures that while statistically significant, fall short of many practitioners' requirements for dedicated strategy deployment. This underscores the efficiency of markets in rapidly incorporating sentiment information into prices.

6.2 Practical Implications

These findings have implications for practitioners considering sentiment-based trading strategies. First, the superior performance of market-level sentiment suggests that traders might benefit from focusing their sentiment analysis efforts on broad market indicators rather than individual stock news. This approach could reduce both computational overhead and the noise inherent in stock-specific sentiment signals.

The revelation about low-coverage stocks performing better presents an interesting opportunity, but one that comes with implementation challenges. While these stocks might offer better predictive potential, they often have lower liquidity and higher trading costs. Many theoretically profitable trades became impractical once realistic trading constraints were applied. This creates a frustrating paradox where the most predictable opportunities are often the hardest to exploit profitably. The results also suggest that traditional approaches to news source selection might need revision. Rather than focusing exclusively on premium news sources, practitioners might benefit from incorporating a broader range of sources, paying particular attention to those demonstrating consistent sentiment profiles over time. The key seems to be temporal stability rather than perceived source quality.

Computationally, the experience suggests that the returns to complexity in sentiment analysis have diminishing returns. While sophisticated NLP models like the instruction-pretrained LLaMA 3 8B perform well at sentiment classification, this improved accuracy doesn't necessarily translate into proportionally better trading performance. Simpler models with carefully designed features might offer a better trade-off between computational cost and predictive power.

On the portfolio management front, the findings support a more nuanced approach to sentiment-based trading. Rather than trying to predict individual stock returns directly from sentiment, traders might benefit from using sentiment analysis as a risk management tool or as one component of a broader multi-factor strategy. The hierarchical nature of sentiment signals suggests that sentiment analysis might be more valuable for sector rotation or market timing decisions than for stock-specific trading.

6.3 Limitations and Future Research Extensions

Throughout this research, several significant limitations warranted acknowledgement and suggested promising directions for future investigation. The most fundamental limitation is survivorship bias in the dataset. By focusing on stocks currently trading, companies that were delisted, acquired, or went bankrupt during the study period (2009-2023) were excluded. This likely makes the results appear more favourable than what a trader would have achieved in real-time, as surviving companies tend to be more stable and successful than the average firm. Data quality issues was also a persistent challenge. Despite using multiple new sources, coverage remains inconsistent across both stocks and periods. The quality of sentiment classification, while strong by current standards, still struggles with nuanced financial language and context-dependent meanings. For instance, phrases like "better than expected losses" or "priced in" often carry sentiment implications that even advanced NLP models could miss.

The approach to news timing and market hours introduces another limitation. By using 4:30 PMEST as the cutoff for same-day trading decisions, important after-hours news flow is potentially missed. However, any alternative cutoff time would introduce different biases. This timing challenge reflects a fundamental difficulty in sentiment-based trading: the constant flow of information doesn't align neatly with market trading hours.

The focus on U.S. stocks trading in U.S. markets limits the generalizability of the findings. International markets, with different trading hours and news cycles, might show different sentiment-return relationships. Additionally, the analysis period (2009-2023) encompasses a generally bullish market environment, which might not reflect sentiment dynamics during more prolonged bearish periods.

Future Research Extensions

Several promising research directions emerge from these limitations: First, extending the approach to intraday data could provide valuable insights. By analyzing how sentiment impacts returns at different frequencies (hourly, 15-minute, or even tick-by-tick), the temporal dynamics of sentiment incorporation into prices could be better understood. This would require high-frequency news data with precise timestamps, consideration of market microstructure effects, analysis of how sentiment impact varies during different parts of the trading day, and study of pre-market and after-hours sentiment effects. Sentiment might have different predictive values at different times of day, with potentially stronger signals during the market open and close. This could lead to more targeted trading strategies focused on specific time windows.

Second, exploring different prediction horizons could yield valuable insights. While this research focused on next-day returns, sentiment might better predict returns over different time frames. Very short-term (intraday) returns might show stronger sentiment effects before price adjustment occurs, medium-term (weekly or monthly) predictions might better capture sentiment trends and long-term (quarterly) forecasts might better align with fundamental value realization.

Third, adjusting the hierarchical level of prediction presents another interesting avenue. Instead of predicting individual stock returns, the focus could be placed on industry-level return prediction using aggregated sector sentiment, predicting broad market movements using economy-wide sentiment indicators, developing rotation strategies between sectors based on relative sentiment, and studying sentiment spillover effects between related industries. Initial tests suggest that industry-level prediction might offer better signal-to-noise ratios than individual stock prediction, particularly in sectors with strong inter-company correlations.

Additionally, exploring alternative prediction targets beyond returns could reveal more robust applications of sentiment analysis. While sentiment shows limited predictive power for next-day returns, it might better forecast other market variables. For instance, sentiment extremes or unusual sentiment patterns could help predict future volatility spikes, as extreme sentiment often precedes periods of market uncertainty. Trading volume prediction might prove more tractable than return prediction, as news sentiment often correlates with increased market activity regardless of price direction. Sentiment analysis could also help identify potential earnings surprises by capturing subtle shifts in market expectations before announcements. Additionally, sentiment might better predict abnormal returns rather than absolute returns, and significant sentiment divergences from historical patterns could signal unusual market behaviour. These alternative applications could provide valuable trading signals even when direct return prediction proves challenging.

The rapid evolution of NLP technology also suggests potential improvements in sentiment extraction, such as testing emerging large language models for more nuanced sentiment analysis, developing finance-specific pre-training approaches, exploring multi-modal models that combine text with market data, and investigating attention mechanisms for identifying relevant news.

These extensions could help bridge the gap between statistical predictability and practical trading profitability, while also deepening the understanding of how sentiment information flows through markets.

7 Conclusion

This research demonstrates both the potential and limitations of sentiment-based stock return prediction. While statistically significant predictive signals were found, particularly in market-level sentiment and less-covered stocks, the practical challenges of implementing these insights into profitable trading strategies remain substantial.

Several key conclusions emerge from the analysis. First, the relationship between news sentiment and stock returns appears to be weakening over time, as evidenced by the declining variance in sentiment-return agreement rates from 2009 to 2023. This suggests that markets are becoming more efficient at processing sentiment information, potentially due to the proliferation of algorithmic trading strategies. Second, the hierarchical nature of sentiment signals, with market-level sentiment proving more reliable than stock-specific sentiment, suggests that traditional bottom-up approaches to sentiment analysis might need reconsideration. The success of market-level signals indicates that sentiment might be more valuable for broad market timing than for individual stock selection.

Third, the superior performance of low-coverage stocks reveals an interesting market efficiency, but one that proves difficult to exploit due to liquidity constraints and trading costs. This illustrates a broader theme in the findings: statistical predictability doesn't always translate into practical profitability.

Looking forward, several promising directions for future research emerge. The relationship between news coverage intensity and predictability deserves deeper investigation, particularly in understanding why less-covered stocks show better predictability. The role of retail-oriented news sources in price formation, especially given their surprisingly stable sentiment profiles, merits further study.

Additionally, the interaction between sentiment signals and market microstructure could provide valuable insights. Understanding how sentiment information flows through different market participants and affects trading behaviour might help explain the gaps between statistical predictability and realized returns. While the results might seem discouraging for pure sentiment-based trading strategies, they suggest more promising applications in risk management and portfolio optimization. Sentiment analysis appears more valuable as one component of a broader investment process rather than as a standalone predictor of returns.

In conclusion, while the holy grail of reliable next-day return prediction through sentiment analysis remains elusive, the research reveals valuable insights about market information processing and the practical limitations of sentiment-based trading. These findings should help both researchers and practitioners develop more realistic approaches to incorporating sentiment analysis into investment strategies.

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A Features Implementation Details

A.1 Sentiment Feature Definitions

A.1.1 Time Decay

Reflects how sentiment information loses relevance over time while preserving recent signals. More recent sentiment scores receive higher weights, modeling the natural decay of information importance in financial markets.

$$s_t^{\text{decayed}} = \sum_{j=0}^T w_j s_{t-j}$$

where:

- $T = 42$ days (maximum lookahead)
- $w_j = \alpha^j / \sum_{j=0}^T \alpha^j$ with $\alpha = 0.95$ (decay factor)
- s_{t-j} is the sentiment value j days ago

A.1.2 Data Freshness

Tracks information staleness, helping identify periods where sentiment signals might be less reliable due to dated information. Critical for low-coverage stocks where news can be infrequent.

$$f_t = \begin{cases} t - \max\{k < t : s_k \text{ is non-null}\} & \text{if such } k \text{ exists} \\ T_{\max} & \text{otherwise} \end{cases}$$

where $T_{\max}$ is the maximum allowed gap.

A.1.3 Raw Sentiment Scores

Provides a baseline measure of market sentiment over different time horizons, capturing both immediate market reactions and longer-term sentiment trends.

$$\bar{s}_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} s_{t-i}^{\text{decayed}}$$

where $w \in \{1, 5, 21, 42\}$ days represents immediate, weekly, monthly, and bi-monthly sentiment averages.

A.1.4 Sentiment Momentum

Captures the acceleration or deceleration in sentiment changes, helping identify shifts in market perception that might precede price movements.

$$\bar{s}_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} s_{t-i}^{\text{decayed}}$$

where $w \in \{1, 5, 21, 42\}$ days represents immediate, weekly, monthly, and bi-monthly sentiment averages.

A.1.5 Sentiment Momentum

Captures the acceleration or deceleration in sentiment changes, helping identify shifts in market perception that might precede price movements.

$$\Delta s_t(d) = \bar{s}_t(w_1) - \bar{s}_{t-d}(w_1)$$

where $d \in \{1, 5, 21\}$ days measures short-term, medium-term, and long-term sentiment momentum.

A.1.6 Sentiment Volatility

Measures sentiment stability and uncertainty. High sentiment volatility might indicate conflicting market views or upcoming significant events.

$$\sigma_t(w) = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} \left(s_{t-i}^{\text{decayed}} - \bar{s}_t(w) \right)^2}$$

where $w \in \{21, 42\}$ days captures monthly and bi-monthly sentiment stability.

A.1.7 Sentiment Trend

Identifies directional patterns in sentiment evolution, helping detect sustained shifts in market perception that might not be captured by simple averages.

$$\beta_t(w) = \frac{\sum_{i=0}^{w-1} (i - \bar{i})(s_{t-i}^{\text{decayed}} - \bar{s}_t(w))}{\sum_{i=0}^{w-1} (i - \bar{i})^2}$$

where $w \in \{21, 42\}$ days measures the strength and direction of sentiment trends.

A.1.8 Sentiment Dispersion

Measures the range of sentiment opinions, helping identify periods of market consensus versus disagreement. High dispersion might indicate uncertainty or potential volatility.

$$d_t(w) = \max_{i \in [t-w+1, t]} (s_i^{\text{max}} - s_i^{\text{min}})$$

where $w \in \{5, 21\}$ days captures weekly and monthly opinion spread.

A.1.9 Sentiment-Return Correlation

Measures how well sentiment aligns with price movements, helping identify periods where sentiment signals might have stronger predictive power.

$$\rho_t(w) = \text{corr}(s_{t-i}^{\text{decayed}}, r_{t-i})_{i=0}^{w-1}$$

where $w \in \{21, 42\}$ days examines different timeframes of sentiment-price relationships.

A.1.10 Lagged Sentiment-Return Correlation

Explores delayed relationships between sentiment and returns, helping identify how long it typically takes for sentiment to be incorporated into prices.

$$\rho_t^l(w) = \text{corr}(s_{t-l-i}^{\text{decayed}}, r_{t-i})_{i=0}^{w-1}$$

where $l \in \{1, 5\}$ days captures different reaction delays.

A.1.11 Sentiment-Volume Interaction

Identifies sentiment signals that coincide with unusual trading activity, potentially indicating stronger market conviction or attention.

$$i_t(w) = \frac{1}{w} \sum_{j=0}^{w-1} s_{t-j}^{\text{decayed}} \cdot (v_{t-j}^{\text{norm}} - 1)$$

where v_t^{norm} is volume normalized by its 42-day average.

A.1.12 Sentiment-Weighted Returns

Measures the historical effectiveness of sentiment signals in predicting returns, giving more weight to periods where strong sentiment aligned with significant price movements.

$$swrr_t(w) = \frac{1}{w} \sum_{j=0}^{w-1} s_{t-j}^{\text{decayed}} \cdot r_{t-j}$$

where $w \in \{5, 21\}$ days examines short and medium-term relationships.

A.1.13 Cross-Level Sentiment Comparisons

Identifies when stock-specific sentiment diverges from broader market or industry sentiment, potentially signaling company-specific developments.

$$\begin{aligned} \Delta s_t^{\text{ind}}(w) &= s_t^{\text{stock}}(w) - \bar{s}_t^{\text{industry}}(w) \\ \Delta s_t^{\text{econ}}(w) &= s_t^{\text{stock}}(w) - \bar{s}_t^{\text{economy}}(w) \end{aligned}$$

where $w \in \{1, 5, 21, 42\}$ days captures divergences across different timeframes.

A.1.14 Relative Sentiment Momentum

Measures whether a stock's sentiment is changing more rapidly than its industry or the broader market, potentially identifying emerging stock-specific trends.

$$\begin{aligned} m_t^{\text{rel, ind}} &= \frac{\Delta s_t^{\text{stock}}(5)}{\Delta s_t^{\text{industry}}(5)} \\ m_t^{\text{rel, econ}} &= \frac{\Delta s_t^{\text{stock}}(5)}{\Delta s_t^{\text{economy}}(5)} \end{aligned}$$

A.1.15 Relative Article Count

Identifies unusual levels of media attention, which might signal important events or changing market focus. Particularly useful for detecting emerging news-driven trading opportunities.

$$c_t^{\text{rel}}(w) = \frac{N_t}{N_t(w)}$$

where $w \in \{5, 21\}$ days compares current coverage to recent historical levels.

A.2 Market Features

Market features complement sentiment indicators through price-based metrics, relative performance measures, and cross-asset relationships.

A.2.1 Simple Returns

Measures price changes to capture price momentum. Different horizons help distinguish between short-term reversals and longer-term trends.

$$r_t(\tau) = \frac{P_t}{P_{t-\tau}} - 1$$

where:

- $r_t(\tau)$ is the return at time t over period τ
- P_t is the price at time t
- $P_{t-\tau}$ is the price τ days ago - $\tau \in \{1, 5, 10, 21, 63\}$ days

A.2.2 Excess Returns

Isolates stock-specific performance by removing market and industry effects. Helps identify outperforming/underperforming stocks.

$$\begin{aligned} er_t^{\text{mkt}}(\tau) &= r_t^{\text{stock}}(\tau) - r_t^{\text{mkt}}(\tau) \\ er_t^{\text{ind}}(\tau) &= r_t^{\text{stock}}(\tau) - r_t^{\text{ind}}(\tau) \end{aligned}$$

where:

- $er_t^{\text{mkt}}(\tau)$ is excess return over market
- $er_t^{\text{ind}}(\tau)$ is excess return over industry
- $r_t^{\text{stock}}(\tau)$ is stock return
- $r_t^{\text{mkt}}(\tau)$ is market return
- $r_t^{\text{ind}}(\tau)$ is industry return

A.2.3 Moving Averages

Smooths noisy price data to reveal underlying trends and provide reference levels for support/resistance.

$$MA_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} r_{t-i}$$

where:

- $MA_t(w)$ is moving average at time t
- r_{t-i} is return i days ago
- $w \in \{5, 21, 63, 126\}$ days

A.2.4 Volatility

Quantifies price uncertainty and risk. Different windows capture short-term turbulence versus long-term risk patterns.

$$\sigma_t(w) = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r}_t)^2}$$

where:

- $\sigma_t(w)$ is volatility at time t over window w
- r_{t-i} is return i days ago
- $\bar{r}_t$ is mean return over window w
- $w \in \{21, 63, 252\}$ days

Volatility Ratio

Compares recent to historical volatility to identify regime changes.

$$vol_ratio_t = \frac{\sigma_t(21)}{\sigma_t(252)}$$

where:

- $\sigma_t(21)$ is 21-day volatility
- $\sigma_t(252)$ is 252-day volatility

Ratio > 1 suggests increasing market stress; < 1 indicates calming markets

A.2.5 Trend

Estimates the strength and direction of systematic price movement, filtering out noise.

$$\text{trend}_t(w) = \beta_1 \text{ from } r_t = \beta_0 + \beta_1 t + \epsilon_t$$

where:

- β_1 is the slope coefficient (trend strength)
- β_0 is the intercept
- t is time index
- ϵ_t is the error term
- $w \in \{21, 63\}$ days

A.2.6 Relative Performance

Identifies stocks outperforming or under performing their benchmarks, suggesting potential momentum continuation.

$$\begin{aligned} rp_t^{mkt} &= MA_t(21) - MA_t^{mkt}(21) \\ rp_t^{ind} &= MA_t(21) - MA_t^{ind}(21) \end{aligned}$$

where:

- $MA_t(21)$ is stock's 21-day moving average
- $MA_t^{mkt}(21)$ is market's moving average
- $MA_t^{ind}(21)$ is industry's moving average

A.2.7 Return Dispersion

Measures price range to gauge market uncertainty and identify potential reversal zones.

$$\text{disp}_t(w) = \max_{i \in [t-w+1, t]} r_i - \min_{i \in [t-w+1, t]} r_i$$

where:

- $\text{disp}_t(w)$ is return dispersion
- r_i is daily return
- $w \in \{21, 63\}$ days

A.2.8 Higher Moments

Captures non-normal characteristics of returns that might signal future price movements.

Skewness

$$\text{skew}_t = \frac{1}{21} \sum_{i=0}^{20} \left(\frac{r_{t-i} - \bar{r}_t}{\sigma_t} \right)^3$$

where:

- skew_t is skewness at time t
- $\bar{r}_t$ is mean return
- σ_t is standard deviation

Negative skewness indicates higher downside risk; positive suggests upside potential.

Kurtosis

$$\text{kurt}_t = \frac{1}{21} \sum_{i=0}^{20} \left(\frac{r_{t-i} - \bar{r}_t}{\sigma_t} \right)^4$$

where:

- Terms as defined in skewness

Higher kurtosis suggests more frequent extreme returns, potential for surprises.

A.2.9 Volume-Weighted Returns

Adjusts return importance based on trading activity, emphasizing price moves with stronger conviction.

$$vwr_t(w) = \frac{\sum_{i=0}^{w-1} r_{t-i} V_{t-i}}{\sum_{i=0}^{w-1} V_{t-i}}$$

where:

- r_{t-i} is return i days ago
- V_{t-i} is trading volume
- $w \in \{5, 21\}$ days

Higher volume-weighted returns suggest stronger, more reliable price moves.

A.2.10 Hurst Exponent

Measures long-term memory in returns, distinguishing between trending and mean-reverting behavior.

$$H_t(w) = \frac{1}{2} \beta_1 \text{ from } \log(\tau) = \beta_0 + \beta_1 \log(\text{lags } s)$$

where:

- $\tau(\ell)$ is range of returns over lag ℓ
- $w \in \{21, 63\}$ days

$H > 0.5$ suggests trending behavior; H close to 0.5 indicates Random Walk; $H < 0.5$ indicates mean reversion.

A.2.11 Volume Features

Relative Volume

Identifies unusual trading activity relative to recent history.

$$rv_t = \frac{V_t}{\frac{1}{21} \sum_{i=0}^{20} V_{t-i}}$$

where:

- V_t is current volume
- Denominator is average volume

Volume Trend

Captures systematic changes in market participation.

$$\text{vtrend}_t = \beta_1 \text{ from } V_t = \beta_0 + \beta_1 t + \epsilon_t$$

where:

- Terms as defined in price trend

Volume Volatility

Measures consistency of trading activity.

$$\text{vol}_t = \sqrt{\frac{1}{20} \sum_{i=0}^{20} (V_{t-i} - \bar{V}_t)^2}$$

where:

- V_{t-i} is volume
- $\bar{V}_t$ is mean volume

Relative Volume Ratio

Compares recent to historical trading activity.

$$\text{rvr}_t(w) = \frac{\frac{1}{w} \sum_{i=0}^{w-1} V_{t-i}}{\frac{1}{21} \sum_{i=0}^{20} V_{t-i}}$$

where:

- Terms as in relative volume
- $w \in \{5, 21\}$ days

B Technical Implementation Details

B.1 Signal Generation

B.1.1 Combined Prediction Score:

$$\Delta_{i,t+1} = \begin{cases} w_h \frac{\Delta^h}{r_{i,t+1}} + w_l \frac{\Delta^l}{r_{i,t+1}}, & \text{if } N_i \in [T_h, T_l] \\ \frac{\Delta^h}{r_{i,t+1}}, & \text{if } N_i > T_l \\ \frac{\Delta^l}{r_{i,t+1}}, & \text{if } N_i < T_h \end{cases}$$

where:

- $\frac{\Delta^h}{r_{i,t+1}}$ is high-coverage model prediction
- $\frac{\Delta^l}{r_{i,t+1}}$ is low-coverage model prediction
- N_i is average annual article count
- $T_h = 50, T_l = 168$ are coverage thresholds
- Weights w_h, w_l calculated as:

$$w_h = \frac{N_i - T_h}{T_l - T_h}, \quad w_l = 1 - w_h$$

B.2 Portfolio Optimization

B.2.1 Objective Function:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T \hat{\mathbf{r}} - \lambda \mathbf{w}^T \Sigma \mathbf{w}$$

where:

- $\mathbf{w}$ is vector of portfolio weights
- $\mathbf{r}$ is vector of predicted returns
- Σ is return covariance matrix
- $\lambda = 1.0$ is risk aversion parameter

B.2.2 Constraints:

$$\begin{aligned} \sum_i w_i &= 1 && \text{(fully invested)} \\ 0 \leq w_i &\leq 0.1 && \text{(position limits)} \\ \sum_i w_i P_i V_i / \text{ADV}_i &\leq C && \text{(capacity constraint)} \end{aligned}$$

where:

- P_i is stock price
- V_i is position volume
- ADV_i is average daily volume
- $C = 10^9$ is base capacity

B.2.3 Position Size Calculation:

$$\text{size}_i = \frac{w_i V_t}{P_i}$$

where:

- w_i is optimized weight for stock i
- V_t is total portfolio value at time t
- P_i is current price of stock i

B.2.4 Portfolio Value Evolution:

$$V_t = V_{t-1} \left(1 + \mathbf{w}_{t-1}^T \mathbf{r}_t - \kappa \sum_i |\Delta w_{i,t}| \right)$$

where:

- $\mathbf{r}_t$ is vector of stock returns
- $\kappa = 0.001$ is transaction cost rate
- $\Delta w_{i,t}$ is change in position weight from $t - 1$ to t .

This framework is implemented with daily rebalancing, accounting for transaction costs and market impact in the backtesting engine. All parameters are calibrated based on empirical analysis of the training period.